

**SUPPLEMENT:
THE DEVELOPMENT AND CULTURAL TRANSMISSION OF RISK
ATTITUDES**

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In this document, we present supplementary plots for the study “The Development and Cultural Transmission of Risk Attitudes”.

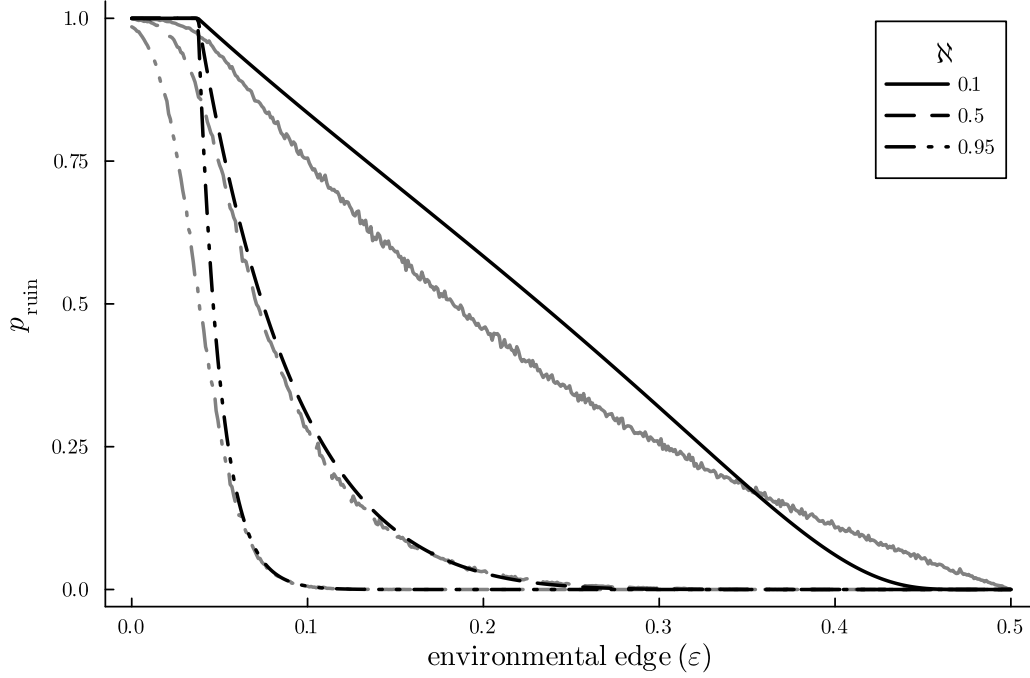


FIGURE S1. **The probability of ruin.** Plotted lines showing the simulated probability of ruin (grey) versus the one predicted by the approximation in the main text (black). $10e^4$ agents were simulated gambling for 1000 time periods using stakes of $s = 0.15$, and the number of ruined agents at the end of these periods is used to calculate the proportion of ruined agents, which is an estimator of the probability of ruin.

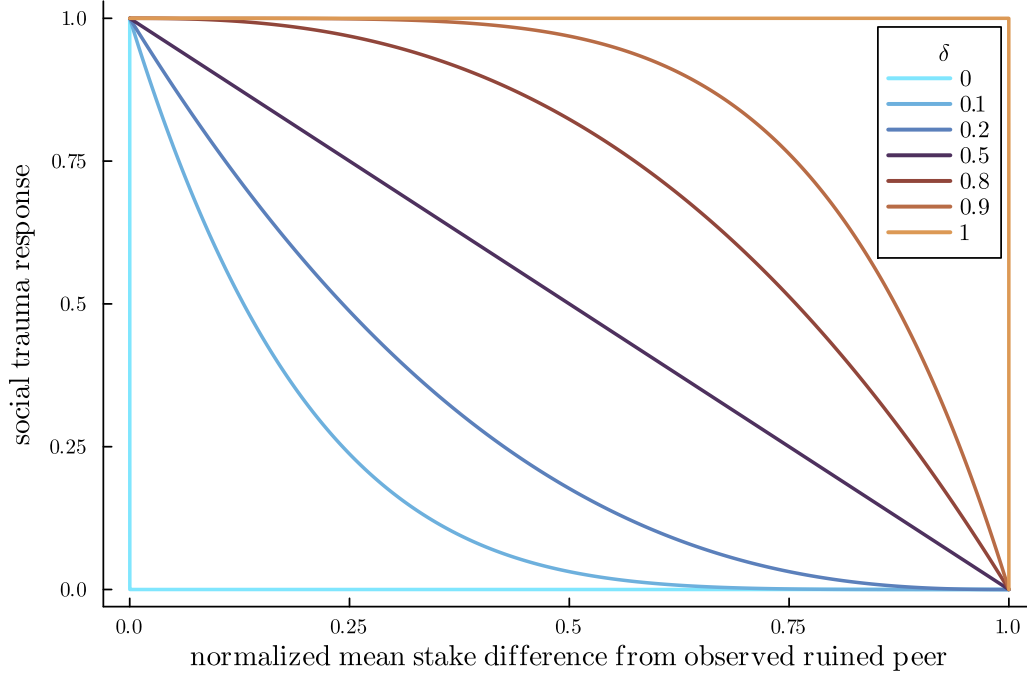


FIGURE S2. **The social trauma function.** Plots of the social trauma function under different sensitivity values. At $\delta = 0$, the social trauma function becomes a Dirac delta function centered at 0. As sensitivity increases, the function exhibit convex decay, with decreasing curvature until it reaches $\delta = 0.5$, the case of linear decay. Above this value, decay becomes concave, with values across the range becoming closer to 1 as sensitivity increase. At $\delta = 1$, the social trauma function equals 1 everywhere in its domain except for when the (normalized) observed difference between ego and the observed ruined peer is 1 (maximum possible difference), at which point the function outputs 0.

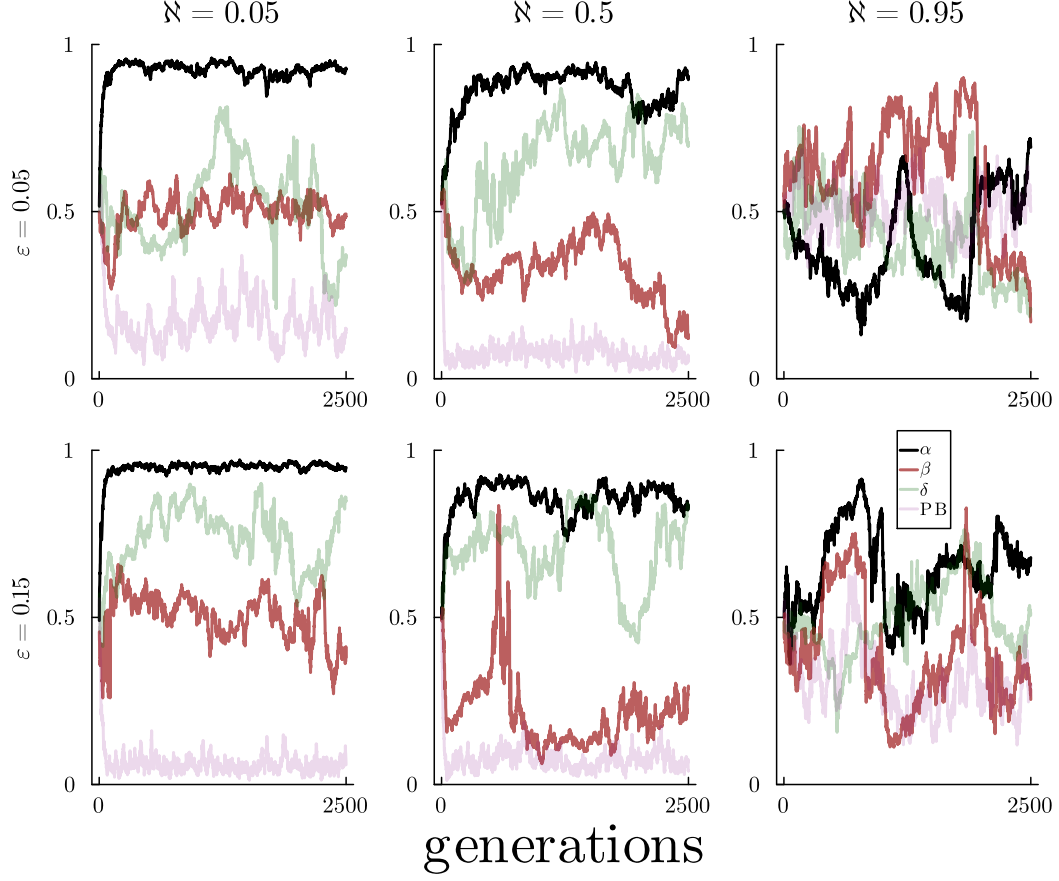


FIGURE S3. **Time dynamics of the elder influence model.** Evolution of learning strategies under different environmental edges and wealth buffers. This plot showcases how increasing wealth buffers decreases the constraints to which learning strategies become subject, allowing them to drift when buffers are high.

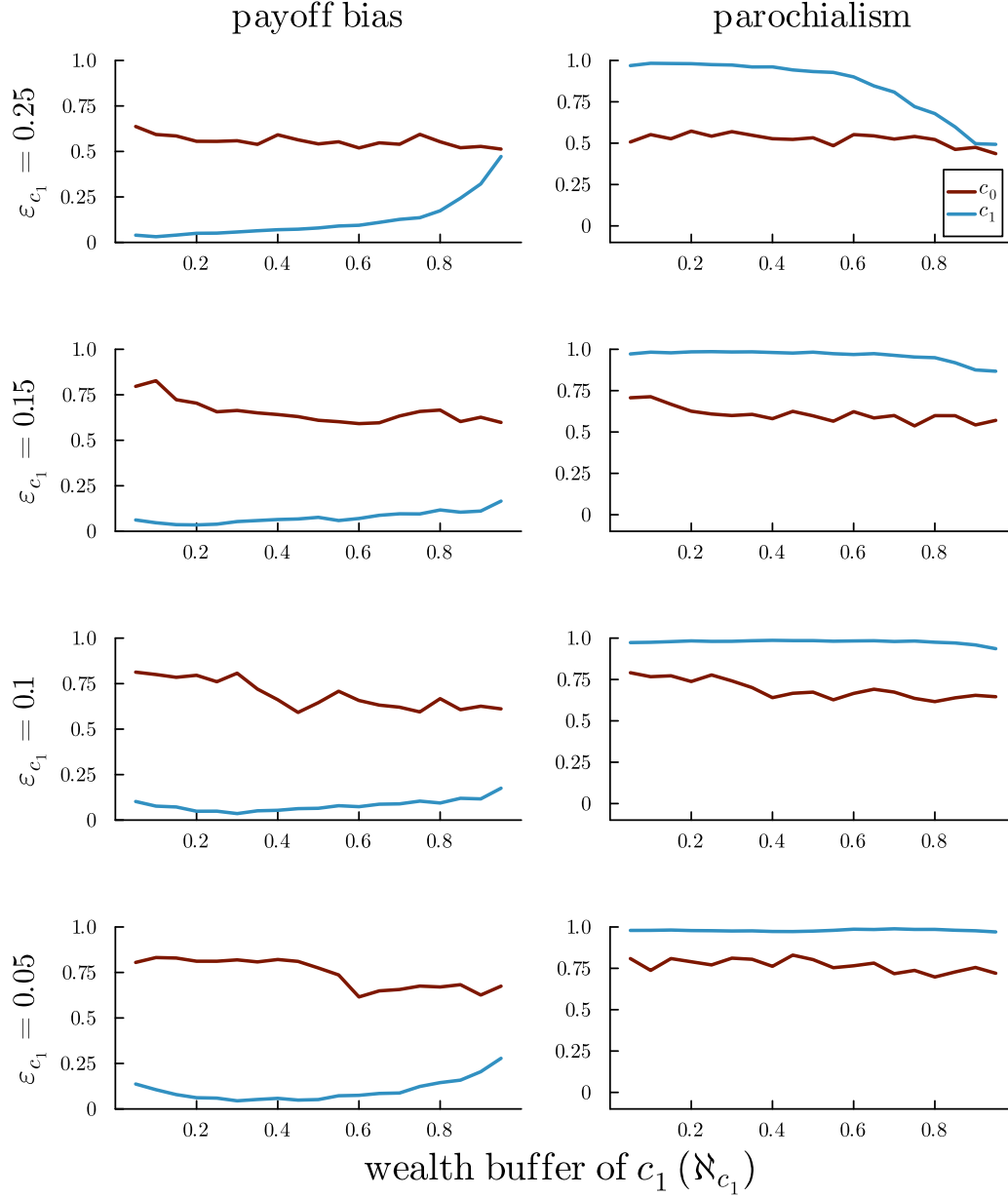


FIGURE S4. Payoff bias and parochialism under different conditions of inequality. Here we vary not only the working class' wealth buffer but also their environmental edge, showing the robustness of the patterns explored in the main text. As the environmental edge of the working class ϵ_{c_1} decreases, not only does the payoff bias of the bourgeoisie increase but so does their explicit parochialism. Since too much payoff bias can lead to agents incurring costs, the bourgeoisie find that it is less costly to evolve an explicit parochial bias for within-class learning. The wealth buffer and environmental edge of the bourgeois class are fixed at $\epsilon_{c_0} = 0.25$; $N_{c_0} = 0.95$.